

Cinema Analytics Data Warehouse

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Project scope and scale

A PostgreSQL data warehouse integrating IMDb and Kaggle movie data for dimensional analysis and OLAP queries.

3

layers: reconciled, warehouse, OLAP

45,431

movies in the reconciled layer

268,615

fact rows in the warehouse

7

OLAP queries; 3 use window functions

Source data and integration problem

Two movie datasets capture complementary aspects of the same domain; the first design decision is how to integrate them without losing source-specific meaning.

IMDb data

- Movies are identified by the unique `tconst` key (e.g. `tt0111161`).
- Ratings, votes, people, crew and principal roles.
- **Strong coverage** for movie identity and audience evaluation; used as the reference for ratings and roles.

Kaggle movie metadata

- TMDb identifiers and production metadata.
- Budget, revenue, countries, languages and companies.
- **Useful for financial and production-oriented analysis**; used to complete what IMDb does not provide.

Without reconciliation, ratings and financials could not be analyzed together.

IDs first; matching only as fallback

39,091 rows resolve from a valid source IMDb ID; exact and fuzzy title matching are controlled fallbacks, and every outcome is audited.

1

Normalize source ID
Canonicalize IMDb ID syntax and distinguish valid IDs from missing or invalid values.

2

Source-ID first
Accept IDs found in eligible IMDb; quarantine valid source IDs that are unavailable.

3

Controlled fallback
Only without a valid ID, require a unique exact title-year match, then a unique same-year fuzzy best.

4

Audit every row
Record method, status and reason; never replace a valid unavailable ID heuristically.

OUTCOME	RECORDS
Accepted mappings	39,096
Quarantined	6,327
Unmatched	10

Audit status governs cross-source identity, not film retention: the loader keeps 45,431 titled Kaggle films, while 39,096 carry an accepted IMDb mapping.

Reconciled schema in third normal form

A normalized hub integrates identity, financials, ratings and production roles while preserving source relationships before warehouse denormalization.

Core entity

The `film` table stores the unified movie identity with IMDb and TMDB references when available.

1

Canonical relations

M:N bridges preserve roles and production data; original language is recovered and Science Fiction becomes Sci-Fi.

2

Integrity controls

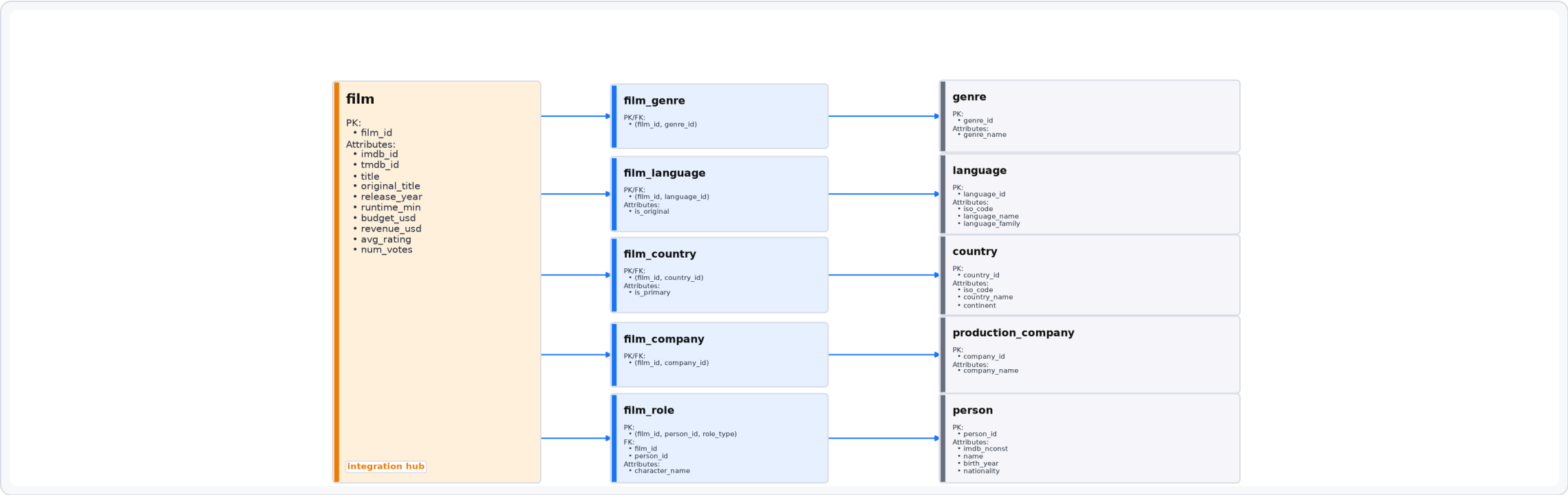
DDL constraints and ETL checks reject invalid keys, role values and source mappings.

3

TABLE	ROWS	ROLE IN THE MODEL
film	45,431	Cleaned Kaggle films, enriched where an IMDb mapping is accepted.
person	183,082	Only people referenced by relevant roles of loaded films.
film_role	474,054	Movie-person-role relationships.
film_genre	100,068	Movie-to-canonical-genre relationships.
film_language	59,847	Spoken and explicitly recovered original-language links.

ER diagram: 3NF reconciled layer

The separate ER view makes the normalized identity hub and its M:N relationship tables explicit.



Central **film** hub with IMDb + TMDb identifiers.

M:N bridges preserve genres, languages, countries, companies and roles.

Fact granularity and dimensions

The DFM fixes the analytical grain before the physical star is built; that choice governs every OLAP aggregation.

Choose analysis axes

1

- **Film** is degenerate; **Time** is year → decade → era.
- **Genre**: genre → genre group.
- **ProductionContext** branches to company and country → continent.
- **Director** is flat; language → language family. Both are optional.

Define the fact grain

2

- **Grain**: film × genre × production context × director × original language.
- Genre, production context and director are multiple dimensions.
- One film → multiple fact rows.

Attach movie measures

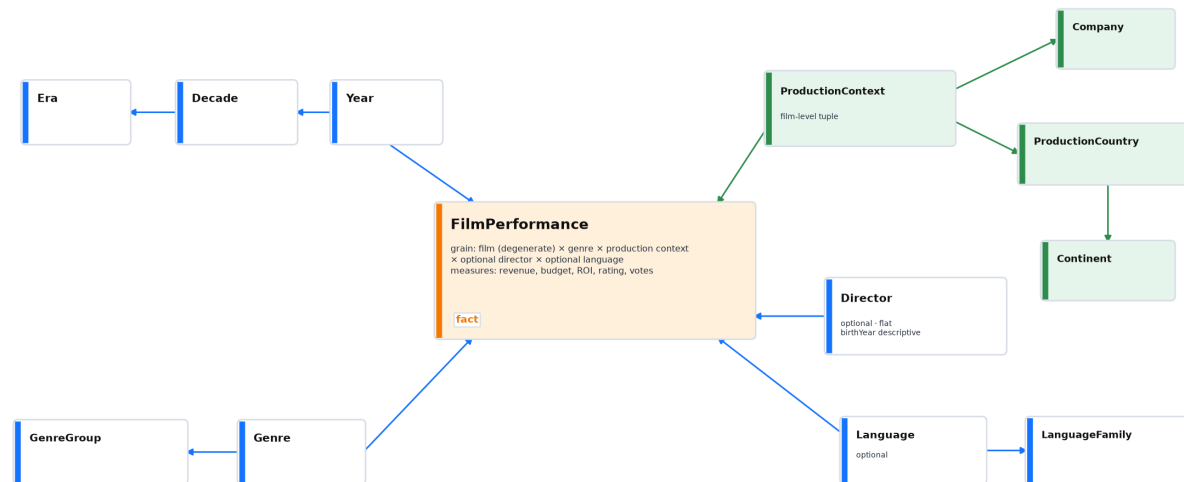
3

- Revenue, budget and ROI from Kaggle.
- Average rating and vote counts from IMDb.
- Deduplicate Film before SUM / AVG; never SUM ROI or rating.

Genre, ProductionContext and Director create fan-out. Director and original language are optional; original language is single and does not multiply rows.

DFM diagram: fact grain and hierarchies

The conceptual model separates the fact, aggregation semantics and analysis coordinates before physical implementation.



ProductionContext branches independently to **Company** and **ProductionCountry** → **Continent**; it never asserts company domicile.

Star schema implementation

Denormalized dimensions keep roll-up, drill-down, slice and dice paths readable around `fact_film_performance`.

Star schema

Five denormalized dimensions around the fact.



Short, readable OLAP join paths.

1

ETL eligibility

Year 1900–2030, at least one genre and one resolvable production company.



33,418 / 45,431 films pass the filter (73.6%).



73.6% fact coverage; 26.4% excluded only by ETL policy.

2

Optional axes

Production context may have no country; Director and Language FKs are nullable.

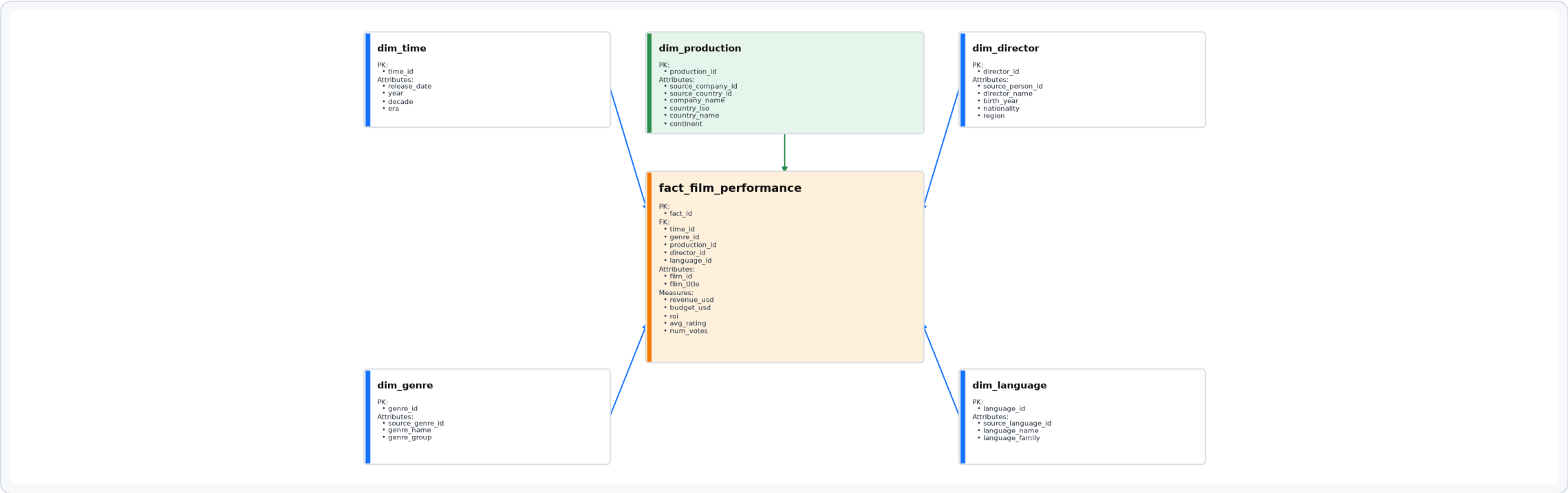


Missing optional data never excludes an eligible film.

3

Star schema: OLAP-ready implementation

The physical warehouse places the film-performance fact at the center of five analytical dimensions.



268,615 fact rows, 33,418 distinct films.

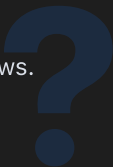
5 dimension tables plus degenerate Film. DDL constraints + ETL preflight/checks: local/row-level integrity.

48,609 / 49,150 production rows have a continent; manual delivery-snapshot checks: cross-row measure consistency + fact coverage.

Deduplication prevents inflated totals

Why the bias appears

- Genres, company-country contexts and directors are M:N — they fan-out fact rows.
- Original language is single and optional, so it does **not** create fan-out.
- Revenue and budget are **movie-level** measures.
- Summing across fan-out rows **inflates** those values.



How analytical validity is preserved

- Use a CTE to deduplicate at the target dimensional level.
- Aggregate only **after deduplication**.
- Apply consistently: results must reflect real data, not modeling artifacts.



EXAMPLE	WITHOUT DEDUPLICATION	WITH DEDUPLICATION
One film: Furious 7	Its revenue is repeated in every genre × company-country × director row.	Collapse the repeated rows into one contribution at the target grain.
Genre–decade result: Action, 2010s	A direct SUM adds those repeated revenues, inflating the group total.	Keep one row per film, decade and genre group, then aggregate.

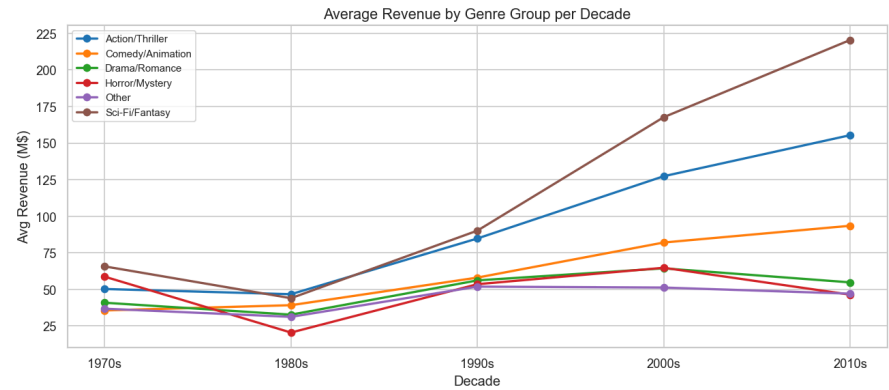
Seven queries cover the required OLAP patterns

Four classical OLAP queries and three window-function analyses.

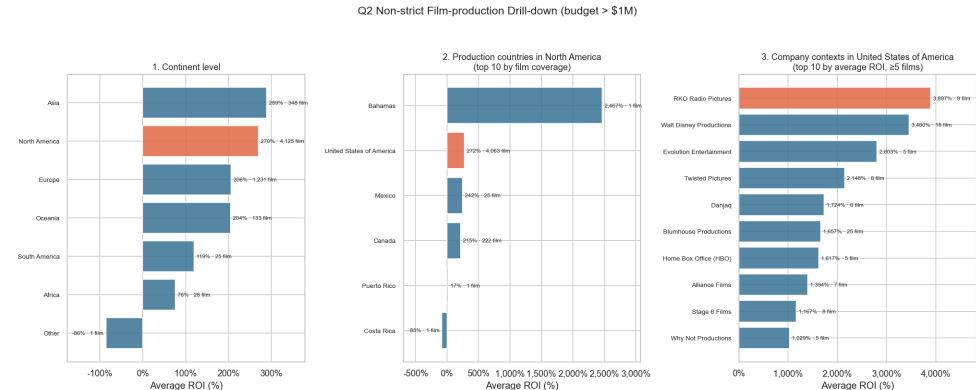
QUERY	OPERATION	ANALYTICAL QUESTION
Classical OLAP operations		
Q1	Roll-up	Revenue by genre group and decade
Q2	Drill-down + slice	ROI from continent to company context
Q3	Dice	Ratings by era and language family
Q4	Slice + CTE	ROI by studio size
Window-function queries		
Q5	NTILE + percentiles	ROI across budget quartiles
Q6	SUM OVER	Cumulative revenue by genre
Q7	RANK	Top-rated films by decade

Visual outputs for Q1–Q4

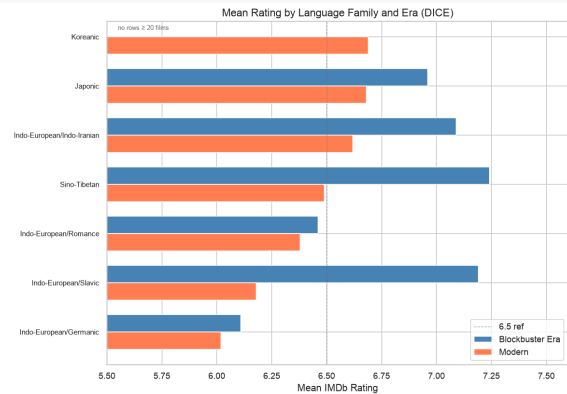
The first four plots connect roll-up, drill-down, dice and slice operations to the current canonical CSV exports.



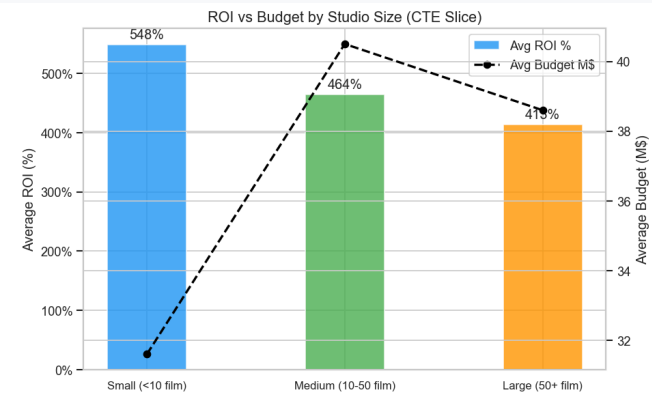
Q1 • Revenue by genre group and decade



Q2 • Production drill-down



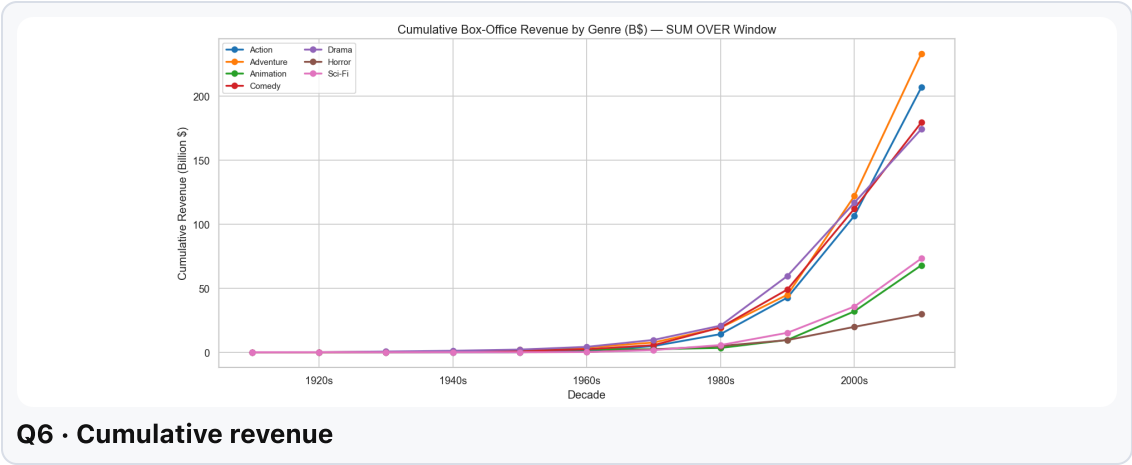
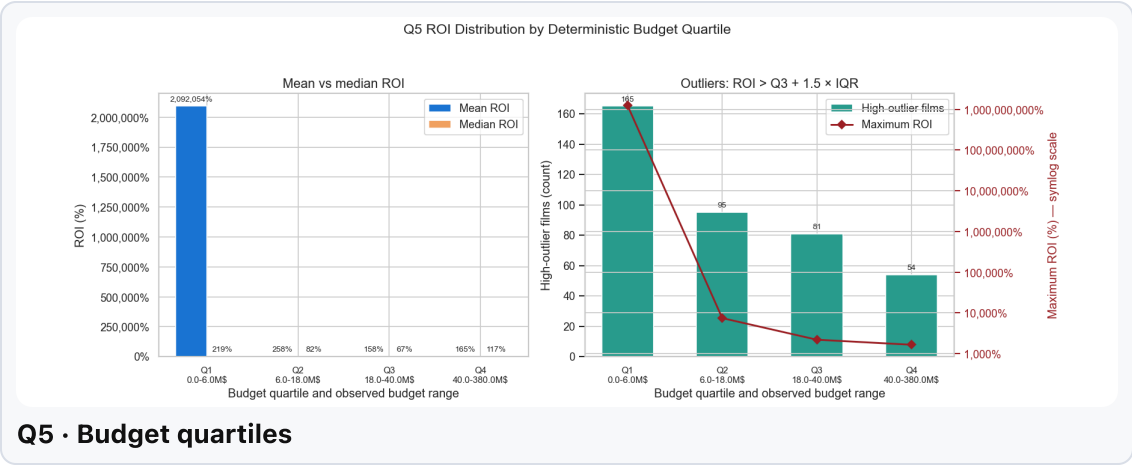
Q3 • Rating by era and language family



Q4 • ROI by studio size

Window-function outputs for Q5–Q7

The remaining plots show NTILE budget bands, independent cumulative SUM OVER series and RANK by decade.



Top-5 Films by IMDb Rating per Decade (RANK() OVER PARTITION BY decade)

Rank	1950s	1960s	1970s	1980s	1990s	2000s	2010s	2020s
#1	12 Angry Men (9.0 ★, 989k v)	The Good, the Bad and the Ugly (8.8 ★, 893k v)	The Godfather (9.2 ★, 2234k v)	The Marathon Family (8.8 ★, 18k v)	The Shawshank Redemption (9.3 ★, 3202k v)	The Dark Knight (9.1 ★, 3183k v)	O.J.: Made in America (8.9 ★, 24k v)	Avatar: The Way of Water (7.5 ★, 634k v)
#2	Seven Samurai (8.6 ★, 403k v)	The Human Condition III: A Soldier's Prayer (8.6 ★, 9k v)	The Godfather Part II (9.0 ★, 1499k v)	The Moromete Family (8.8 ★, 5k v)	Schindler's List (9.0 ★, 1590k v)	The Lord of the Rings: The Return of the King (9.0 ★, 2172k v)	Inception (8.8 ★, 2829k v)	Bad Boys for Life (6.5 ★, 198k v)
#3	The Counterfeit Coin (8.6 ★, 3k v)	World Gone Mad (8.8 ★, 3k v)	Tosun Pasha (8.9 ★, 26k v)	The Kingdom of Diamonds (8.8 ★, 3k v)	Sandesham (9.0 ★, 6k v)	The Lord of the Rings: The Fellowship of the Ring (8.9 ★, 2214k v)	The Phantom of the Opera at the Royal Albert Hall (8.8 ★, 11k v)	The King's Daughter (5.2 ★, 7k v)
#4	The Human Condition I: No Greater Love (8.5 ★, 11k v)	Harakiri (8.6 ★, 92k v)	The Message (8.9 ★, 12k v)	Star Wars: Episode V - The Empire Strikes Back (8.7 ★, 1510k v)	Fight Club (8.8 ★, 2623k v)	The Lord of the Rings: The Two Towers (8.8 ★, 1964k v)	Natsamrat (8.8 ★, 8k v)	The Prey: Legend of Kamoctus (3.6 ★, 1k v)
#5	The Human Condition II: Road to Eternity (8.5 ★, 9k v)	Psycho (8.5 ★, 788k v)	The Chaos Class Is Waking Up (8.7 ★, 22k v)	Stop Making Sense (8.7 ★, 24k v)	Forrest Gump (8.8 ★, 2506k v)	Before Flying Back to Earth (8.7 ★, 1k v)	Interstellar (8.7 ★, 2550k v)	—

Q7 · Top films by decade

What the model supports — and where it stops

Stable invariants protect the model; documented source limits define the boundary of interpretation.

What is demonstrated

- Reconciled, warehouse and OLAP layers run in PostgreSQL.
- Stable integrity checks block invalid states.
- Canonical exports connect every query to its visual output.

Documented limitations

- Director nationality is unavailable in IMDb.
- Production geography is film context, not company domicile.
- Budget coverage limits the financial subset.

Reliable analysis starts before the query

Reconcile identity

Join complementary sources through auditable IDs.

Fix the grain

Define what one fact row represents before aggregation.

Deduplicate first

Aggregate movie-level measures only at the target analytical level.

Summary

The project demonstrates a complete analytical pipeline — two heterogeneous sources, a reconciled integration layer, a dimensional star schema, and seven OLAP queries over the current fact snapshot.

thank you